
EE/CprE/SE 491 WEEKLY REPORT 1

9/14 – 9/18 Group: 28

Project title: Biomimetic Prosthesis with Precise Recreation of Hand Movement

Client &/Advisor: Erik Hrysyk, Meng Lu

Team Members/Role:

Reid Post – Powertrain Researcher

Redwan Elshafiey – Sensors and Control Researcher

Logan Ward – EMG Researcher

Adriana Cedillo Sanchez – Anatomy Researcher

Jacob Page – EMG and Materials Researcher

Erik Hrysyk – EMG/powertrain Researcher

Quinn Snyder – Materials/Joints Researcher

○ **Weekly Summary**

This week the group met in order to set up expectations for the year. This involved setting up weekly meetings, as well as assigning general roles for research for each member moving forward in the next week. The discussion was largely focused on the general expectations that everyone has for the project, and what type of work that they would like to do. It was decided that for now, each person will be given the title of researcher, as we need to work to define the scope of the project with research. In time, each person will be given a more permanent title after addressing how each person works together and what their proficiencies and work style are.

○ **Past week accomplishments**

N/A (project kickoff, team contract was completed)

○ **Pending issues** (If applicable: Were there any unexpected complications? Please elaborate.)

N/A, no pending issues

○ **Individual contributions**

<u>NAME</u>	<u>Individual Contributions</u> <i>(Quick list of contributions. This should be short.)</i>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Logan Ward	Brainstorming, group organization, start on EMG research	6	6
Reid Post	Brainstorming, Group organization, start on powertrain research	6	6
Redwan Elshafiey	Brainstorming, group organization, start on sensors research	6	6
Erik Hrysyk	Brainstorming, Group planning, role selection, start on EMG research	6	6
Jacob Page	Brainstorming, Group Organization, role selection, Resource Identification, Starting on EMG Research	6	6
Adriana Cedillo	Brainstorming, research	6	6
Quinn Snyder	Brainstorming, group organization, start on materials research	6	6

○ **Comments and extended discussion** *(Optional)*

- *This will be the future home of the technical blocks for our project. This is intended to provide a visible location for the problems that each member is having, and hopefully others can see this and provide support*

○ **Plans for the upcoming week**

- Logan – Research EMG in terms of signal processing/controls. Also look into potential machine learning applications.
- Reid – Research different strategies for moving joints, specifically looking into tradeoffs between servos, pneumatics, motors, and also brainstorm other possibilities
- Erik – Research EMG signal processing, test EMG sensors, familiarize myself with lab space, start 3D modeling, look into amputee hand/arm structure
- Adriana- Research actuators, anatomy of the hand and amputees
- Quinn – Research different materials strategies for prosthetics
- Redwan – Research sensor communication and controls
- Jacob – Research EMG signal processing, Test EMG sensors, research potential movement options.

- **Summary of weekly advisor meeting**

First meeting on Monday with advisor, and this will be a recurring weekly meeting

If we need help contact ZZY97@iastate.edu EE Grad Student

Grading criteria

Each weekly report is worth 10 points. Scores will be awarded as follows:

- **8 – 10:** Progress for your project seems to be suitable. Documentation and hours reported by team members are adequate.
- **6 – 8:** There is scope of improvement both in your report and your project progress. Can consult with instructor/TA after class for further inputs.
- **< 6:** Please talk to instructors/TA after class hours about any difficulties that you/your team is facing.

Each weekly report should be unique in that they have a unique set of supporting details for your contributions. So please do not just copy your reports from the previous week. In addition, please avoid any personal pronouns (he, she, I, you). Try to keep your reports as neat as possible.

Advisor Meeting Notes 9/21:

- General introductions
- Talked about some general new courses and technologies in BME (learned about robotic arm)
- Brought up the problem with the current lack of feedback (specifically for the case of if the arm is picking up a soft object)
- Brought up the robotic arm in the MRC, and how that could link to this project. This arm leaves the controller open and could help to keep us from writing the source code from the bottom.
- ROS (robotic operating system) is an open source way that could reduce the amount of code we need to write if our controllers are compatible with it
- Our advisor is more focused on the feedback part of it (how would the operator understand this feedback pressure)
- Optical mouse computer mouse could be used to know the pressure on the fingers
- It sounds like in 2-4 weeks we want to look around at the industry and then get together an actual proposal for what we would like the scope of the project to be
- Because we would like it to be wearable, it is hard to make it testable, and because it is taking human data, we need to apply to IRB to make sure that we can legally use the human biometric data
- Communication with hand, Raspberry Pi won't have enough IO ports
 - May need to use arduino or make our own board

- SparkFUN is a website that would likely have the biometric sensors that we would need
- The machine learning side of it may have a problem with the I/O from an arduino because it will likely need a computer
- Discussed being able to use the 3D printer in the lab for free as well as using any extra electronics laying around
- Grad Student Email: zzy97@iastate.edu, He can help with using/finding lab equipment
- BME students have a class teaching kit has EMG + data acquisition device
- There are open sourced models, for the EMG models for the arm possibly. This can help.
- In order to determine how many motors we need, we need to determine how reliably we can read based off the forearm with the EMG
- There is an opensource pretrained EMG model, don't build our own
- Look into EMG sensors to see how many muscles we can isolate and then decide how
- EMG is a weak analog signal can be measured with any data acquisition software
- <https://keys.ece.iastate.edu/> we should all get access to the lab

Deliverables:

- Provide a preliminary timeline for the project
- By the end of the semester, have a prototype for each of the sensors
- Lock in the scope of what we (react within a certain time threshold, basic functionality, hold a cup etc, general hand symbols,) NOTE: we want all of these to have a number, number of gestures, certain
- Physical Hand team log in and gain a little familiarity with SolidWorks for CAD purposes
- Set up GitHub page for all data aq, signal analysis, machine learning, etc.
- Apply to the IRB for waiver and permissions
- EMG team, look at hardware, how to acquire data, and then general machine learning data
- Check for university Codex subscription